

Repository Status Wave Track Synthesis

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Summary

The repository is in an active but stable post- Wave 2C state. No training campaign is currently prepared or running, and doc/running/active_training_campaign.yaml has no protected files.

The current scalar program winner is

Surface	Run	Family	Test MAE [deg]	Test RMSE [deg]
Bw	te_periodic_gru_sequence_remote_Bw	periodic_gru_sequence_bw	0.002344	0.002747

That scalar winner is not the whole decision anymore. Commit b73220679410276246421b7e2832d8878cff90a0 moved promotion toward full Track 2 curve-following quality.

Current Track 2 decision

- accepted paper-derived baselines remain paper_retuned_best_Fw and paper_retuned_best_Bw ;
- tree remains the strongest repository-owned static baseline;
- Wave 2B is the strongest repository-owned neural branch, while Wave 2C is verified only as an exploratory residual harmonic temporal baseline.

Repository State

The program has closed the following major surfaces:

Surface	Current State	Operational Meaning
Track 1	closed	faithful full-bank RCIM paper reproduction surface
Wave 1	closed	structured static repository baselines
Track 2	closed through Wave 2C	official offline curve-verification matrix
Wave 2	completed	first temporal sequence baselines
Wave 2B	completed	harmonic temporal hybrids, current strongest neural branch
Wave 2C	completed	residual harmonic temporal hybrids, exploratory baseline

The repository now has 53 implemented and benchmarked family surfaces in the master summary. The next active work should not be another blind family addition. The current focus is standardizing curve-first reranking on the expanded Track 2 metric surface.

Operationally, the repository is ready for decision-oriented analysis rather than another bookkeeping pass. The available evidence now covers scalar leaderboards, official curve reports, direction-aware comparisons, and deployment plausibility for the strongest branches.

Wave 1 Static Baselines

Wave 1 remains the closed structured-static baseline wave. It covers direction-separated `global`, `Fw`, and `Bw` surfaces for tree, feedforward, periodic MLP, harmonic regression, and residual harmonic MLP families.

The scalar HPO closeout leader is

Rank	Family	Scope	Engine	Test MAE [deg]	Test RMSE [deg]
1	<code>tree_fw</code>	<code>forward</code>	<code>bounded_grid</code>	0.002743	0.003409
2	<code>tree</code>	<code>global</code>	<code>bounded_grid</code>	0.002782	0.003520
3	<code>tree_bw</code>	<code>backward</code>	<code>bounded_grid</code>	0.002954	0.003749

The post-closeout harmonic tracking campaigns improved the harmonic-specific families but did not displace the tree baseline as the strongest Wave 1 scalar and Track 2 static family.

Best current harmonic-oriented Wave 1 scalar results:

Family Surface	Best Run	Harmonic Basis	Test MAE [deg]
<code>harmonic_regression_fw</code>	<code>te_harmonic_dense360_tracking_Fw</code>	<code>dense 0..360</code>	0.002916
<code>residual_harmonic_mlp_bw</code>	<code>te_residual_harmonic_rcim_sparse_tracking_Bw</code>	<code>sparse RCIM</code>	0.003042
<code>periodic_mlp_fw</code>	<code>te_periodic_mlp_dense240_tracking_Fw</code>	<code>dense 0..240</code>	0.003055
<code>residual_harmonic_mlp_fw</code>	<code>te_residual_harmonic_rcim_sparse_tracking_Fw</code>	<code>sparse RCIM</code>	0.003089

Interpretation

- `dense` harmonics can help simple harmonic regression in forward-only scalar training;
- `sparse RCIM` remains strong for residual harmonic MLP surfaces;
- `periodic_mlp_harmonic_*` is a report/candidate label for explicit-harmonic `periodic_mlp` candidates, not a separate architecture family;
- `tree` remains the strongest repository-owned static Track 2 baseline.

Wave 2 Temporal Sequence Models

Wave 2 introduced temporal sequence baselines

- `temporal_convolution`;
- `gru_sequence`;
- `lstm_sequence`.

The entry campaign completed all 9 planned runs. Its scalar campaign winner was:

Run	Family	Scope	Test MAE [deg]	Test RMSE [deg]
<code>te_gru_sequence_remote_Fw</code>	<code>gru_sequence_fw</code>	<code>Fw</code>	0.003333	0.003881

The best global and backward temporal-entry surfaces were:

Surface	Run	Family	Test MAE [deg]
<code>global</code>	<code>te_lstm_sequence_remote_global</code>	<code>lstm_sequence</code>	0.003482
<code>Bw</code>	<code>te_lstm_sequence_remote_Bw</code>	<code>lstm_sequence_bw</code>	0.003557

Track 2 verified these models as useful exploratory temporal baselines, but plain temporal recurrence did not improve enough over the static tree and paper-derived reference candidates.

Wave 2B Harmonic Temporal Hybrids

Wave 2B added explicit periodic harmonic features to temporal convolution, GRU, and LSTM sequence windows. The first tier used the sparse RCIM harmonic list:

text [0, 1, 3, 39, 40, 78, 81, 156, 162, 240]

The campaign completed all 9 planned runs. Its scalar winner is also the current program scalar winner:

Run	Family	Scope	Test MAE [deg]	Test RMSE [deg]
te_periodic_gru_sequence_remote_Bw	periodic_gru_sequence_bw	Bw	0.002344	0.002747

The strongest bidirectional neural candidate is

Run	Family	Scope	Test MAE [deg]	Test RMSE [deg]
te_periodic_gru_sequence_remote_global	periodic_gru_sequence	global	0.002681	0.002971

Track 2 confirms the same qualitative result. The official visual and metric reports identify:

Candidate	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error [%]
periodic_gru_sequence_Bw	Bw	0.002392	0.002639	5.466
periodic_gru_sequence_global	global	0.002704	0.002949	6.139
periodic_lstm_sequence_global	global	0.002707	0.002958	6.120

Interpretation

- adding explicit sparse harmonic features to recurrent temporal models is the most successful neural move so far;
- GRU and LSTM are nearly tied in global scalar and curve metrics, but the backward-only periodic GRU is the clearest winner;
- the result is still less deployability-transparent than static tree or harmonic models, so the next decision must be curve-first and TwinCAT-aware.

Wave 2C Residual Harmonic Temporal Hybrids

Wave 2C tested a stronger decomposition

- structured harmonic branch for the base TE shape;
- recurrent temporal branch for sequence-local residual correction;
- final prediction as the sum of the two branches.

The campaign tested 18 runs:

- residual_harmonic_gru_sequence ;
- residual_harmonic_lstm_sequence ;
- global, Fw, and Bw surfaces;
- sparse RCIM, dense 0..240, and dense 0..360 harmonic banks.

The scalar campaign winner was:

Run	Family	Basis	Scope	Test MAE [deg]	Test RMSE [deg]
te_residual_harmonic_gru_sequence_remote_Fw_sparse_rcim	residual_harmonic_gru_sequence_fw_sparse_rcim	sparse RCIM	Fw	0.003200	0.003635

The scalar leaderboard showed dense variants close to sparse variants in some forward-only pointwise metrics.

Track 2 changed the interpretation: when the models are judged on full curves, dense residual harmonic banks are clearly weaker.

Strongest Wave 2C Track 2 candidates

Scope	Candidate	Basis	Curve MAE [deg]	Curve RMSE [deg]	Mean Error [%]
Fw	residual_harmonic_gru_sequence_sparse_rcim_Fw	sparse RCIM	0.003194	0.003499	7.083
Bw	residual_harmonic_lstm_sequence_sparse_rcim_Bw	sparse RCIM	0.003440	0.003793	7.510
global	residual_harmonic_lstm_sequence_sparse_rcim_global	sparse RCIM	0.003368	0.003719	7.409

Dense Wave 2C Track 2 examples

Scope	Candidate	Basis	Curve MAE [deg]	Mean Error [%]
Fw	residual_harmonic_gru_sequence_dense240_Fw	dense 0..240	0.006983	15.722
Fw	residual_harmonic_gru_sequence_dense360_Fw	dense 0..360	0.007869	17.740
Bw	residual_harmonic_lstm_sequence_dense240_Bw	dense 0..240	0.007367	16.660
Bw	residual_harmonic_lstm_sequence_dense360_Bw	dense 0..360	0.010268	23.355
global	residual_harmonic_lstm_sequence_dense240_global	dense 0..240	0.006419	14.460
global	residual_harmonic_lstm_sequence_dense360_global	dense 0..360	0.008810	19.916

Interpretation

- sparse RCIM is the correct wave 2C harmonic tier;
- dense residual harmonic expansion overfits or destabilizes the curve surface;
- residual decomposition is useful as a diagnostic branch, but it does not beat the simpler wave 2B periodic recurrent formulation;
- wave 2C should stay closed as a verified exploratory baseline.

Track 2 Current Result

The official Track 2 package now contains 111 candidates. It evaluates direction-valid held-out TE curves and reports curve-level metrics separately from scalar training metrics.

Current official leaders

Scope	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]
forward overall	rcim_retuned_GBM19_Fw	0.001089	0.001299	2.372
backward overall	periodic_gru_sequence_Bw	0.002392	0.002639	5.466
paper-derived forward	paper_retuned_best_Fw	0.001839	0.002041	4.109
paper-derived backward	paper_retuned_best_Bw	0.003675	0.004284	7.572
repository static backward	tree_Bw	0.003258	0.003651	7.051
repository global static	tree_global	0.003144	0.003533	6.854
repository global neural	periodic_gru_sequence_global	0.002704	0.002949	6.139

The important split is

- the recovered/retuned paper reference still owns the best forward curve result;
- wave 2B owns the best repository neural backward and global neural result;
- wave 1 tree remains the most robust repository-owned static baseline;
- wave 2C adds evidence about harmonic-bank choice, not a new promotion.

Harmonic Bank Conclusions

The harmonic-bank evidence is direction- and family-dependent. There is no single universal winner across every algorithm.

Context	Best Observed Choice	Evidence
Wave 1 harmonic regression, Fw	dense 0..360	test_mae = 0.002916
Wave 1 periodic MLP, Fw	dense 0..240	test_mae = 0.003055
Wave 1 residual harmonic MLP, Bw	sparse RCIM	test_mae = 0.003042
Wave 2B periodic recurrent	sparse RCIM feature tier	periodic_gru_sequence_Bw , test_mae = 0.002344
Wave 2C residual harmonic temporal	sparse RCIM	best scalar and all best Track 2 surfaces
Wave 2C dense residual temporal	dense 0..240 or 0..360	poor Track 2 curve metrics

Practical conclusion

- dense harmonic banks are useful as targeted stress tests and can help some static or low-capacity harmonic models;
- sparse RCIM harmonics are the better default for neural hybrid and residual temporal work;
- dense 0..240 and 0..360 should not be promoted into new temporal residual campaigns without a curve-first reason.

Perspective Shift From Commit b73220679410276246421b7e2832d8878cff90a0

Commit b73220679410276246421b7e2832d8878cff90a0 is titled

text Define curve-first TE training strategy

Its change of vision is important. Before that commit, the repository could be read as mostly scalar-registry driven: a lower test_mae moved the program winner. After that commit, scalar error remains a sanity gate, but not the final promotion rule for TE compensation.

The new interpretation is

1. TE compensation is a continuous curve-following problem.
2. Full held-out curves are evaluation units, not future inputs supplied to the runtime model.
3. Future deployed models must remain causal: current operating state, optional short past history, or causal derived features only.
4. Track 2 curve metrics and visual overlays become the promotion surface.
5. Scalar winner, curve-first winner, and deployment-ready candidate should be reported separately until Track 3 online evidence exists. This is a useful correction. It prevents the next work from chasing a small pointwise MAE gain that may smooth out oscillations, shift harmonic phase, or fail under repeated-revolution compensation.

Future Development Plan

Step 1: Track 2B Curve-First Reranking

Open the already planned `Track 2B Curve-First Reranking` branch before any new training campaign.

Deliverables

- expanded per-curve metric CSV;
- direction-separated reranking for `Wave 1` , `Wave 2` , `Wave 2B` , and `Wave 2C` ;
- harmonic amplitude and harmonic phase diagnostics on the sparse `RCIM` harmonic set;
- P95 and worst-condition percentage-error tables;
- updated visual overlays if the screened candidate set changes;
- master-summary update separating scalar best from curve-first best.

Step 2: Decide Whether To Retrain Or Promote

If reranking confirms that `Wave 2B` periodic recurrent models are also the best curve-first neural branch, keep them as the neural reference and improve checkpoint selection before changing families.

If a static or harmonic model is better curve-first despite weaker scalar metrics, open a compact retraining branch around that family rather than launching a broad new model wave.

Step 3: Add Curve-First Checkpoint Selection

Update neural training infrastructure so checkpoint selection can monitor a validation curve metric, for example:

```
text val_curve_mean_percentage_error_pct
```

Tie-breakers should include curve P95 error, harmonic phase error, then scalar `val_mae` .

Step 4: Add Curve-Aware Losses Only After Metrics Stabilize

Do not immediately train with complex curve losses. First standardize the curve metrics. Then test a small composite loss:

```
text pointwise_normalized_mse + curve_mae_term + slope_or_derivative_term +  
selected_harmonic_amplitude_term + selected_harmonic_phase_term
```

Soft-DTW or other warp-tolerant objectives should remain secondary ablations, because physical angular phase matters for compensation.

Step 5: Preserve A TwinCAT-Friendly Path

The final target is not only an offline benchmark. The selected path must stay credible for TwinCAT/TestRig compensation.

Preferred future candidates

- sparse RCIM harmonic or periodic models with explicit intermediate quantities;
- curve-first selected periodic GRU or LSTM only if deployment cost and inspectability are acceptable;
- residual harmonic MLP or structured harmonic variants with harmonic-space diagnostics;
- later Track 3 online compensation tests for uncompensated versus compensated TE_{RMS} and TE_{max} . Deferred exploratory candidates such as lightweight transformers, state-space sequence models, neural ODEs, and kernel/Gaussian-process baselines should stay low priority until Track 2B clarifies what the existing candidates already do on full curves.

Bottom Line

The work is no longer in a raw model-search phase. The repository now has a complete scalar and Track 2 evidence surface through Wave 2C.

The best scalar neural result is Wave 2B `periodic_gru_sequence_Bw`. The best repository static baseline remains `tree`. The best forward curve result is still paper-derived retuned `GBM`. The most important technical next step is not another large campaign, but a curve-first reranking and promotion policy that decides which existing candidates are actually suitable for continuous TE compensation.